

FinPredict

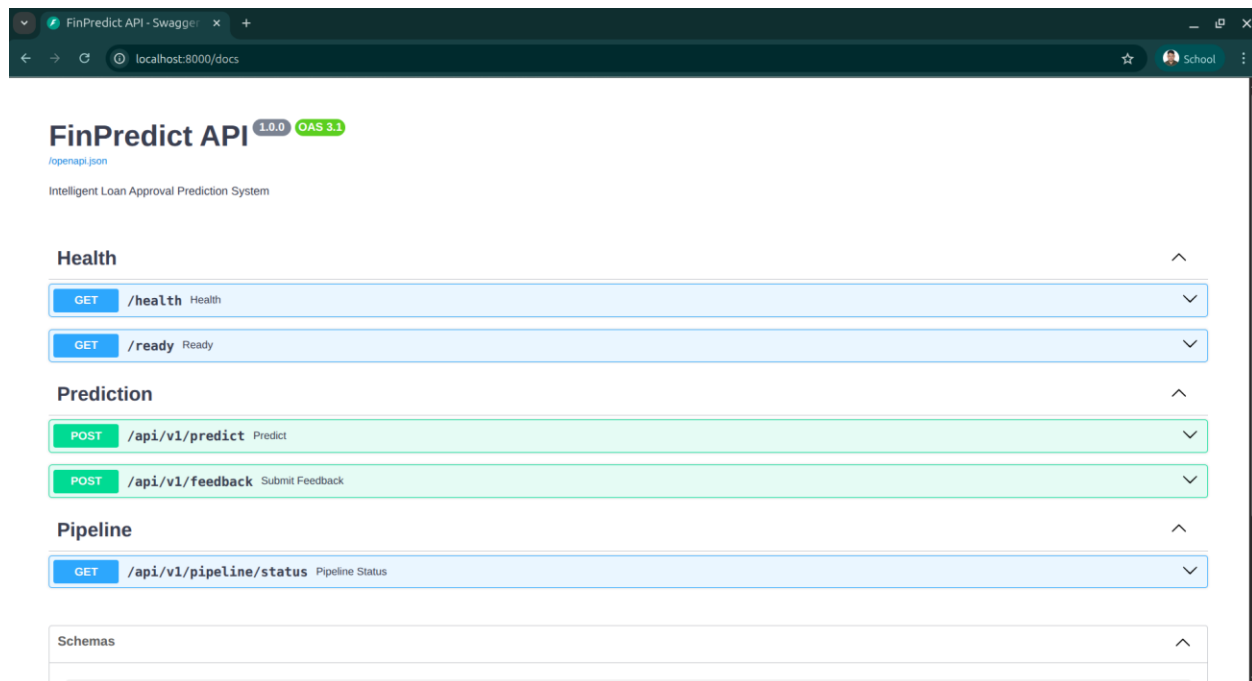
Low-Level Design Document (LLD)

API Endpoint Definitions and I/O Specifications

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1. API Overview

Property	Value
Base URL	http://localhost:8000
Protocol	HTTP/1.1
Content Type	application/json
Authentication	None (open access on internal network)
Framework	FastAPI 0.104.1 with Pydantic v2 validation
Documentation	http://localhost:8000/docs (Swagger UI)
CORS Origins	http://localhost:3002, http://127.0.0.1:3002



2. Endpoint Definitions

2.1 GET /health

Description: Liveness probe. Confirms the server process is running. Used by Docker healthcheck.

Property	Value
Method	GET
URL	/health
Authentication	None
Request Body	None

Response — 200 OK

```
{ "status": "healthy" }
```

2.2 GET /ready

Description: Readiness probe. Confirms the ML model is loaded and the system is ready to accept prediction requests.

Property	Value
Method	GET
URL	/ready
Authentication	None
Request Body	None

Response — 200 OK

```
{ "status": "ready", "model": "FinPredict_XGBoost", "version": "1.0.0" }
```

2.3 POST /api/v1/predict

Description: Core prediction endpoint. Accepts 24 applicant fields, runs preprocessing, and returns a loan approval prediction with probability and risk level.

Property	Value
Method	POST
URL	/api/v1/predict
Content-Type	application/json

Authentication	None
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Request Body Schema

Field	Type	Constraints	Example
gender	string	Enum: Male, Female	Male
age	integer	$18 \leq \text{age} \leq 100$	35
marital_status	string	Enum: Married, Single, Divorced	Married
dependents	integer	$0 \leq \text{dependents} \leq 10$	1
education	string	Enum: Graduate, High School, Postgraduate	Graduate
employment_status	string	Enum: Employed, Self-Employed, Unemployed	Employed
occupation_type	string	Enum: Professional, Salaried, Freelancer, Business	Salaried
residential_status	string	Enum: Own, Rent, Other	Own
city_town	string	Enum: Urban, Suburban, Rural	Urban
annual_income	float	> 0	85000
monthly_expenses	float	> 0	2500
credit_score	integer	$300 \leq \text{score} \leq 850$	720
existing_loans	integer	≥ 0	1
total_existing_loan_amount	float	≥ 0	15000
outstanding_debt	float	≥ 0	8000
loan_history	integer	0 or 1 (0=No, 1=Yes)	0
loan_amount_requested	float	> 0	20000
loan_term	integer	> 0 (months)	120
loan_purpose	string	Enum: Home, Personal, Vehicle, Education	Home
interest_rate	float	> 0	8.5
loan_type	string	Enum: Secured, Unsecured	Secured
co_applicant	string	Enum: Yes, No	Yes
bank_account_history	integer	≥ 0 (years)	5

transaction_frequency	integer	>= 0 (per month)	15
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Response Body Schema — 200 OK

Field	Type	Description	Example
loan_approved	boolean	true = Approved, false = Rejected	true
approval_probability	float	Model confidence 0.0 to 1.0	0.8732
risk_level	string	Low (>0.75) / Medium (>0.5) / High	Low
message	string	Human-readable outcome	Loan Approved
credit_score_impact	string	Assessment of credit score factor	Strong positive factor
recommendation	string	Personalised advice	Your application meets our criteria.

Error Responses

Status Code	When	Response Body
422 Unprocessable Entity	Missing or invalid field values	{ detail: [{ loc, msg, type }] }
500 Internal Server Error	Model inference failure	{ detail: 'error message' }

2.4 POST /api/v1/feedback

Description: Feedback loop endpoint. Accepts ground truth labels for previous predictions to enable real-world performance monitoring.

Property	Value
Method	POST
URL	/api/v1/feedback
Content-Type	application/json
Authentication	None

Query Parameters

Parameter	Type	Description
applicant_id	string	Reference ID for the prediction
predicted	integer	Model prediction (0 or 1)

actual_outcome	integer	Real-world outcome (0 or 1)
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Response — 200 OK

```
{ "status": "feedback logged", "correct": true }
```

2.5 GET /api/v1/pipeline/status

Description: Returns the current status of all 7 Airflow pipeline tasks. Used by the Pipeline Monitor page in the frontend.

Property	Value
Method	GET
URL	/api/v1/pipeline/status
Authentication	None
Request Body	None

Response — 200 OK

```
{ "status": "active",
  "last_run": "2026-04-22T04:18:05 UTC",
  "tasks": {
    "data_ingestion": "success",
    "data_validation": "success",
    "data_cleaning": "success",
    "feature_engineering": "success",
    "encode_and_scale": "success",
    "train_test_split": "success",
    "trigger_training": "success"
  }
}
```

2.6 GET /metrics

Description: Prometheus metrics endpoint. Exposes all custom metrics in Prometheus text format. Scraped by Prometheus every 15 seconds.

Property	Value
Method	GET
URL	/metrics
Response Format	Prometheus text format

Authentication	None
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Exposed Metrics

Metric Name	Type	Labels	Description
finpredict_prediction_latency_seconds	Histogram	none	End-to-end prediction latency
finpredict_predictions_total	Counter	status: success/error	Total prediction requests
finpredict_http_requests_total	Counter	method, endpoint	Total HTTP requests
finpredict_data_drift_score	Gauge	none	Current drift score 0.0-1.0

3. Module-Level Design

3.1 Backend Module Structure

Module	File	Responsibility
Main App	api/main.py	FastAPI app creation, middleware setup, router registration
Predict Route	api/routes/predict.py	POST /predict handler, drift calculation, metrics
Health Route	api/routes/health.py	GET /health and GET /ready handlers
Pipeline Route	api/routes/pipeline.py	GET /pipeline/status handler
Feedback Route	api/routes/predict.py	POST /feedback handler
Request Schema	api/schemas/request.py	LoanApplication Pydantic model (24 fields)
Response Schema	api/schemas/response.py	PredictionResponse Pydantic model (6 fields)
CORS Middleware	api/middleware/cors.py	Cross-origin request configuration
Model Loader	model/loader.py	Load model from MLflow registry
Predictor	model/predictor.py	LoanPredictor class, singleton pattern
Preprocessor	model/preprocessor.py	Feature encoding and engineering
Metrics	monitoring/metrics.py	Prometheus Counter, Histogram, Gauge setup
Config	config.py	Environment variable loading

3.2 Preprocessing Pipeline (model/preprocessor.py)

The preprocessor applies these transformations in order to match the training pipeline exactly:

Step	Columns Affected	Transformation
1. Binary Encoding	Gender, Loan_Type, Co_Applicant	Male→1/Female→0, Secured→1/Unsecured→0, Yes→1/No→0
2. Ordinal Encoding	Education	High School→0, Graduate→1, Postgraduate→2
3. Feature Engineering	New columns	Debt_to_Income, Loan_to_Income, Expense_to_Income ratios
4. One-Hot Encoding	Marital_Status (3), Employment_Status (3), Occupation_Type (4), Residential_Status (3), City_Town (3), Loan_Purpose (4)	pd.get_dummies — 20 new columns
5. Column Ordering	All 41 columns	Strict FINAL_COLUMNS order matching training data